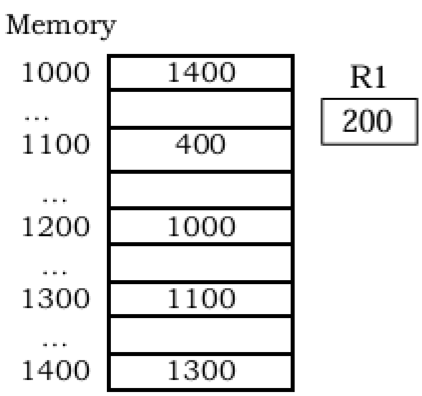
Introduction to Computer Architecture

Sheet #8

Exercises:

1. What is the difference between using direct and indirect addressing? Give an example.

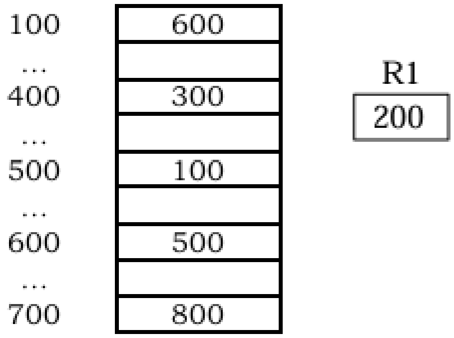
2. Suppose we have the instruction Load 1000. Given that memory and register R1 contain the values below:



Assuming R1 is implied in the indexed addressing mode, determine the actual value loaded into the accumulator and fill in the table below:

Mode	Value Loaded into the AC
Immediate	
Direct	
Indirect	
Indexed	

3. Suppose we have the instruction Load 500. Given that memory and register R1 contain the values below:



Assuming R1 is implied in the indexed addressing mode, determine the actual value loaded into the accumulator and fill in the table below:

Mode	**Value Loaded into the AC**
Immediate	
Direct	
Indirect	
Indexed	
Register	
Register Indirect	
Based Addressing	

4. A nonpipelined system takes 200ns to process a task. The same task can be processed in a 5-segment pipeline with a clock cycle of 40ns. Determine the speedup ratio of the pipeline for 200 tasks. What is the maximum speedup that could be achieved with the pipeline unit over the nonpipelined unit?

5. A nonpipelined system takes 100ns to process a task. The same task can be processed in a 5-segment pipeline with a clock cycle of 20ns. Determine the speedup ratio of the pipeline for 100 tasks. What is the theoretical speedup that could be achieved with the pipeline system over a nonpipelined system?

6. Write code to implement the expression: A= (B + C) * (D + E) on 3-, 2-, 1- and 0-address machines. In accordance with programming language practice, computing the expression should not change the values of its operands.

7. The memory unit of a computer has 256K words of 32 bits each. The computer has an instruction format with 4 fields: an opcode field; a mode field to specify 1 of 7 addressing modes; a register address field to specify one of 60 registers; and a memory address field. Assume an instruction is 32 bits long. Answer the following:
    a. How large must the mode field be?
    b. How large must the register field be?
    c. How large must the address field be?
    d. How large is the opcode field?